

LLM: Training & Performance

Training: architectureni structure rasi tayyor bulgandan keyingi jarayon. Training jarayoni tez, arzon va minimum hatol bilan bulishi kerak

Training jarayon asosan 3 ta componentdan iborat:

- Hyperparameters
- Stability of training
- Regularization

Hyperparameters

Hyperparameters: model uchun eng yaxshi hyperparameterlarni topib berish jarayon

LLM da muhim hyperparameterlar:

- Learning Rate (LR)
- Batch size
- Optimizer
- Warm_steps
- Weight_decay

Feature	Asosiy task	Froniter modellarda
Base LR	Gradient descent dagi qadam uzunligi	0.0001 ~ 0.003 micro range
LR Scheduler	Yuqoridan pastga yurish qoidasi	Cosine decay: ohirida LR qiymatini 0 ga tushuradi
Global batch size	Bitta qadamda GPU lar jami qancha tokenni bir vaqtda hisoblash qoidasi	GPT-4 da 1M~4M token gacha beriladi
Per-device BS	bitta GPU memory siga sigadigon kichik hajm	2,4,8,16 qilib belgilanadi
Optimizer	Gradient yunalishini va compasni Hisoblaydigon asosiy math motor	AdamW: beta1=0.9, beta2=0.95
Warm-up steps	LR ni birdaniga katta bermasdan, sekin-sekin isitib borish usul	Jami LR ning 1~5% beramiz

Gradient clipping	Gradientlarni kuchini kesib turadi	Max-norm treshold 1.0 beramiz
Weight decay	Overfittingni oldini olish uchun ogirlik berish	Standard qiymat: 0.1 (AdamW)

Training Stability

Training stability: model necha marta training qilingandaham natijasi stabilniy yaxshi bulishi.

muammolar:

- **Exploding Gradients:** kup layerlar va warmup qilinmagan bulsa, kup hato matnlar berilgan bulsa gradient portlaydi
- **Qaerdan bilvolamiz?:** loss kamayishni urniga juda kattayib ketadi. va loss imiz NaN yoki Inf qiymatlarga teng bulib qoladi

Yuqoridagi muammolarni yechish bilan training stability ni oshiramiz:

- **LR Warmup:** LR ni 0 dan boshlab sekin-sekin oshiramiz
- **Gradient Clipping:** har bir gradient uchun treshold quyish. Va agar usha qiymatdan oshib ketsa orqaga qaytadi. Ex) frontier=1

Regularization

Regularization: overfitting ni urniga tushunishni taminlash. Yani, overfittingni oldini olish.

Usullari:

1. **weight decay:** har qadamda NN ni ogirligini 0 ga qarab pasaytirib turadi
2. **dropout:** malum bir oraliqni yoki qismni tashlab yuboradi

LLM da weight decay uzi yetarli buladi. chunki, LLM da overfitting nisbatdan kamroq

Hugging Face

Hugging face platformasi – AI ni GitHub i deb tushunsak buladi. bu platformada turli AI modellar bor. GPT-2 modeliniham uz ichiga oladi. Va barchasi open-source malumot

<https://huggingface.co/> - link

GPT-2 Repository:

- Tokenizer
 - tokenizer.json (tokenizer settings)
 - tokenizer_config.json (token to ID mapping)
 - vocab.json (BPE merge rules)
 - merges.txt (vocabulary and rules)

 - Model Configuration
 - config.json

 - Neural Network
 - model.safetensors
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Bizham model yasayotkanimizda shunday formatdan foydalanamiz.

✂ `pip install transformers sentencepiece` – bu command orqali biz VSCode bilan HuggingFace ni connect qivolamiz